

Project Name: Social Media and Mental Wellbeing

Team Members: Maria Jose Healey, Moonsoo Kim, Jasmine Lau, Kimsean Pen

Final Presentation: [Link](#)

Project URL (final version): [Link](#)

Member Participation:

Team Members	Maria Jose Healey	Moonsoo Kim	Jasmine Lau	Kimsean Pen
Sections	Demographics	Interactions	Platform	Time Spent
Presentation	27.5%	10%	35%	27.5%
Usability Study	30%	10%	30%	30%
EDA	50%	—	—	50%
Website Display	27.5%	10%	35%	27.5%
Proposal	50%	—	—	50%

↓ changes that were made based on usability study ↓

Demographics

In the prototype, the demographics section included only 1 dashboard which consisted of 3 charts displayed in a horizontal layout. The first one was a side by side bars that allowed subjects to observe emotions expressed by social media users based on their gender. This visualization only had an 84.6% success rate at displaying different emotions accurately because its visibility was affected by the location of this graph inside the dashboard. This caused some categories of emotions to not be listed below their representative bars. The second chart in this dashboard was a treemap. This visualization helped users identify the number of social media users under each emotion category. This graph was a minor usability issue, as only 76.9% of users were able to answer a related question correctly. Additionally, some people in the usability study survey pointed out that they were not familiar with this type of chart which made it difficult to navigate through. The third chart was a stacked bars chart that displayed the distribution of emotions based on gender and age. This visualization provided filters for the users to refine their exploration of the graph. However, it was not easily operated by the test subjects, as the majority rated its ease of use lower than the highest score in a 5 rating scale.

The usability study yielded a prioritization list that helped improve all the visualizations. The first dashboard in the demographics section was employed to address some items from this list such as, creating visualizations that are more simple, adding more context, and considering color accessibility.

To indicate the number of social media users that fit into the different categories for the “Gender” variable, a pie chart was utilized. This simple graph demonstrates the percentage and quantity of each gender included in the dataset. Likewise, a box-and-whisker plot provides similar information for the “Age” attribute. This plot shows the spread and range of distribution of the social media users’ age and it also makes it easy to compare the different age groups.

Packed bubbles (circular treemaps) were used to communicate the emotions grouped into clusters. The size of each bubble represents the number of people who manifest such feelings, and the color corresponds to different emotions. In addition, the color blind palette was used in this dashboard and all subsequent dashboards and/or charts.

The 2nd component of the demographics section contains a drop down menu where you can swap between dashboards that hold only one chart. These are displayed separately one at a time in the same container. Additionally, these charts provide interactivity through drop down menus, sliders, and filters. This is the approach taken to take care of other items in the prioritization list such as, the need for displaying each graph on separate dashboards to improve visibility, changing the orientation of bars, and adding drop down menus. The first option on the drop down menu list is an interactive horizontal bars graph. This chart is easier to navigate than the side by side bars and provides a clear view of the distribution of emotions based on gender. The second option is an interactive scatter plot. Unlike the treemap, this graph makes it easier to observe the frequency of occurrence of emotions based on age. The addition of a slider, drop down menu, and the use of different colors for the variety of emotions made this visualization more user-friendly. The interactive horizontal bars chart display as the 3rd option on the drop down menu list makes it easier to read information and compare attributes in contrast to the stacked bars chart. Also, the slider and drop down menu provide a straightforward way to tailor the display of information according to different interests.

Platform

From the results of our usability study's questions specific to the platform dashboard, we found that this dashboard received the fewest comments and 85 - 93% of the respondents answered the questions related to this dashboard correctly. This high accuracy meant that the original iteration of the dashboard was effective in getting information about the dominant emotions experienced on the platforms across to users of our website. As there was not much negative feedback about these visualizations, I decided to just clean them up and take what criticisms were given to the website as a whole and implement changes to meet them.

One of the comments received was the survey participants by and large did not know what a heatmap is, so had to infer to find it, but aside from not knowing what a heat map is, another comment was about the labels being cut off, taking this feedback, I ensured that in our final product, none of the words are cut off and the annotations are clear. Since our users may not know what a heatmap is, I decided to place less emphasis on its interactions and made it reactive and show the specifics of what the users selected in the more familiar chart above it, the proportional bar chart.

Looking back at the initial version of the heatmap, I felt as though the individual cells kind of merged together, since there was no clear delineation between the cells, and that the diverging color encoding did not make sense since the values themselves are not opposite of one another, as a diverging color scale may suggest. The values themselves are the number of people reporting a specific emotion when on a platform. So, in the final product, I ensured that the cells are clearly defined with a grid, and changed the diverging coloring to a sequential color encoding. Overall, the sentiment that we received from the survey was that a more landscape view was preferred, so I took the other half of my dashboard, being the proportional bar chart, and switched the x and y axes. Keeping in mind the comment about labels being cut off, the final product also ensures that labels are clear and the bar chart acts as a filter for the heatmap below since the bar chart is more simple, using the heatmap as a more detailed view into our dataset seems to be a better choice than the previous interaction.

Putting this all together, from the first iteration of this dashboard to the final one, not much in ways of content has changed. The most obvious change is the orientation of the proportional bar chart. I found this switch to be a better use of space, as previously, the amount of space wasted as well as the whitespace left by it, may have contributed to the overall mental load of our users, so the landscape orientation should help to cut down on this. The addition of the heat map grid as well as the sequential color encoding also help to tie this dashboard together to be easy to understand and easy to interact with.

Time Spent

Approximately 76.9% of respondents successfully identified the platform with the highest average time spent. Many users indicated that the dashboard was overwhelming, and missing the ability to switch between different minutes to accurately determine the platform with the highest time spent. To improve the accuracy score of these observations, I transitioned to a treemap to display which platform usage based on time spent. This visualization will allow users to precisely identify Instagram as the platform with the highest time spent based on the color and size of the rectangles. Additionally, the treemap graph is linked with dominant emotion based on daily usage graph, allowing users to explore specific emotions by hovering over the corresponding areas, which dynamically reflects on the social media platform graph below.

To improve clarity, the graph displaying dominant emotions based on daily usage was simplified by switching to a box plot. Previously, users were confused by the varying size of the dots and had difficulty identifying the dominant emotions based on time spent. Additionally, a single data point at 200 minutes plotted for aggression led to incorrect assumption that aggression was the dominant emotion for extended social media use. This confusion arose because the visualization did not adequately represent a distribution of the data where readers are able to read the graph easily. The box plot, by contrast, clearly shows the distribution range and reveals happiness as the dominant emotion for longer time spent on social media users supported by various sample counts. Color encoding was also removed to further reduce visual noise and enhance clarity.

User noted that the scale for “User Engagement by Daily Usage Time” was too high, making visualization and comparison difficult. Additionally, separating different interactions into separate graphs hindered comparison. In response, I combined all three graphs into a single visualization and used different shapes to plot each interaction type. This allows users to easily differentiate between interaction types and make more effective comparisons.

Although not highlighted in the usability study, I decided to rename some of the graphs with more specific and objective titles to clearly convey the main insight to users. The original titles were somewhat wordy and even caused confusion for me, so the new titles aim to provide clearer guidance on the graphs’ content.

Interaction

I received feedback indicating that graphs taking the average for all ages do not drive meaningful insights because social media activity varies significantly across different age groups. A more solid feedback referenced a statistical fact that younger individuals are exposed to more social media activity. Additionally, the stacked bar chart I previously used caused confusion, as it was difficult to discern the average values per dominant emotion due to the size of the bars.

Based on this feedback, I decided to rework the graph by incorporating age as a parameter. This adjustment allows us to see the dominant emotions for each age group and to identify the largest and smallest dominant emotions by their average values within each age. I reworked the visualization using Tableau to ensure consistency with other graphs created by my teammates. Additionally, I updated the colors of the dominant emotions to match the criteria used in other graphs, which also display dominant emotions for different purposes.

An interesting observation emerged after incorporating the age parameter. Initially, this change addressed the feedback regarding the importance of age as a parameter. Intriguingly, I noticed that for the age group 21-23, only four dominant emotions were present. However, after age 24, two additional dominant emotions appeared consistently across all subsequent ages. This insight suggests that emotions such as "aggression" and "boredom" become more prominent as individuals mature, starting from age 24 on average.

In the final visualization, I included both a scatter plot and a bar chart on the same dashboard. These graphs are designed to change dynamically based on different metrics and age groups selected by the user. For easier readability and to illustrate the crucial role of age differences, the scatter plot remains fixed for all ages, while the bar chart adjusts based on the average values for different age groups.

By comparing these two graphs, we can gain a deeper understanding of how social media interactions and dominant emotions vary across different age groups, providing valuable insights into the behavioral patterns of users on social media platforms.

Overall

Given more time, it would have been nice to interlink all our graphs to display various analysis and metrics for each emotion interactively. However, due to time constraints, we focused on creating graphs for our individual sections. Additionally, it would be valuable to collect further demographic data, such as education level, income, as well as data about users' lifestyle factors to visualize if that influences their choice in social media platforms.